

SE3080 – Software Project Management
Assignment 01 – Component Ownership & Workload Division
Group_018 | Elderly Companionship & Micro-Volunteering Platform
Repository: github.com/dilmithp/SE3080-Group18_Assignment01

1. Group Members & Scrum Roles

Student ID	Name	Scrum Role
IT23600416	Wijekoon W M V M	Product Owner + Development Team
IT23543300	Ranketh K A D	Scrum Master + Development Team
IT23534254	Pathirana P P D S S	Development Team
IT23670334	Perera K K L A	Development Team

Roles are fixed across Sprint 0–2. In line with Scrum, the Product Owner and Scrum Master also contribute as developers so that all four members own a substantial implementation component and can demonstrate individual technical contribution at the viva.

2. Technology Stack

Layer	Technology	Justification
Mobile Front-End	Flutter (Dart)	Single codebase for Android & iOS; strong accessibility widget support for elderly users
Backend / Database	Firebase Firestore	Real-time NoSQL DB — suits live session availability and booking status updates
Authentication	Firebase Authentication	Email/phone OTP sign-in; low-friction for elderly users
Cloud Logic	Firebase Cloud Functions	Server-side matching logic, trust-score calculation, scheduled reminders
Notifications	Firebase Cloud Messaging	Push reminders for upcoming sessions; SMS fallback for non-smartphone users
Storage	Firebase Storage	Profile photos and verification document uploads
Version Control	Git + GitHub	Feature-branch workflow with pull requests; commit history evidences individual contribution
Project Management	Jira / Trello	Sprint board, backlog, burndown charts (rubric criterion 5)
Design	Figma	Wireframes, design system and clickable prototype (shared with the UEE module)

3. Component Ownership

The system is divided into four vertical components. Each member owns their component end-to-end — the UI screens, the business logic and the Firestore data model — so that every member has a demonstrable, independently explainable technical contribution as required for the viva.

Member	Owned Component	Scope of Work (UI + Logic + Data)
Pathirana P P D S S (IT23534254)	Component 1: Identity, Access & Trust Management	• Firebase Authentication: email/phone OTP registration and login for both elderly users and volunteers • Role-based access control (elderly / volunteer / admin) with Firestore security rules • Volunteer verification workflow — document upload to Firebase Storage plus admin approval queue

		• Trust score model: Cloud Function computing a score from completed sessions, ratings and verification status • Report / block user and emergency-escalation flow • UI screens: sign-up, login, OTP, verification status, admin approval dashboard
Perera K K L A (IT23670334)	Component 2: User Profiles & Accessibility Framework	• Elderly and volunteer profile data models in Firestore (skills offered, help needed, availability preferences, locality) • Profile creation, editing and photo upload (Firebase Storage) • Accessibility framework as reusable Flutter widgets: dynamic text scaling, high-contrast theme, simplified navigation mode, voice-guided prompts • Shared design system / component library used by all four members (buttons, cards, form fields, typography scale) • Onboarding wizard adapting to user type and accessibility preference • UI screens: onboarding wizard, profile view/edit, accessibility settings
Wijekoon W M V M (IT23600416)	Component 3: Matching & Discovery Engine	• Matching algorithm (Cloud Function) pairing elderly requests with volunteers by locality, skill, availability and trust score • Search and filter interface — browse volunteers or open help requests by category and distance • Ranked recommendation list with match-reason explanation shown to the user • Geolocation handling and distance calculation within a locality radius • Volunteer detail view: profile summary, trust badges, reviews, availability preview • UI screens: discovery/browse, filters, match results, volunteer detail
Ranketh K A D (IT23543300)	Component 4: Scheduling, Sessions & Community Feedback	• Peer-to-peer scheduling: availability calendar, session request, accept/decline, recurring sessions • Booking lifecycle state machine in Firestore (requested → confirmed → completed / cancelled) with conflict handling • Automated reminders via Firebase Cloud Messaging with SMS fallback (scheduled Cloud Function) • Post-session feedback loop: ratings and reviews feeding back into the trust score • Community engagement layer: local activity feed, volunteering streaks and badges • UI screens: calendar, booking request/confirmation, session history, feedback form, activity feed

4. Workload Weighting

Story points estimated using the Fibonacci scale (5, 8, 13) per work type. Totals are balanced within one story point per member across the two development sprints.

Member / Component	UI Screens	Business Logic	Data / Backend	Integration	Total SP
Pathirana (Component 1)	8	13	8	13	42
Perera (Component 2)	8	13	13	8	42
Wijekoon (Component 3)	5	8	13	13	39
Ranketh (Component 4)	5	8	13	13	39

Total planned velocity: 162 story points across Sprint 1 and Sprint 2, averaging 40.5 points per member. Components 1 and 2 carry marginally more foundational work, which is offset by Components 3 and 4 carrying the heavier algorithmic and integration work in Sprint 2.

5. Sprint Delivery Plan

Sprint	Member	Tasks
Sprint 0 Jul 28 – Aug 3 (Setup)	Pathirana	- Initialize Flutter project, folder architecture and CI-friendly repo structure - Create Firebase project; configure Auth, Firestore and Storage - Set up GitHub branching strategy, README, .gitignore, PR template
	Perera	- Build the shared design system in Figma and as Flutter theme/widgets - Define accessibility standards the whole team must follow (min font size, contrast ratio, tap-target size)
	Wijekoon	- Write Project Charter / Product Vision - Draft and prioritize the Initial Product Backlog (epics → user stories → tasks with story-point estimates) - Confirm stakeholder / customer proxy for sprint reviews
	Ranketh	- Set up Jira/Trello board, sprint structure and burndown tracking - Define the team's Definition of Done and ceremony calendar - Draft Sprint 1 and Sprint 2 plans with the Product Owner
Sprint 1 Aug 4 – Aug 14 Goal: a verified user can register, build a profile and be discovered	Pathirana	- Implement registration, login and OTP flows with Firebase Auth - Implement role-based access and Firestore security rules - Build the volunteer verification upload + admin approval screen
	Perera	- Implement profile data models and profile create/edit screens - Ship the accessibility framework widgets (text scaling, contrast, simplified mode) - Deliver the onboarding wizard
	Wijekoon	- Implement browse/search UI with category and distance filters - Implement basic locality-based query against Firestore - Run Sprint 1 review with the stakeholder and capture feedback
	Ranketh	- Facilitate stand-ups, planning and retrospective; maintain the burndown chart - Build the session data model and availability calendar UI shell - QA the onboarding and auth flows; log defects on the board
Sprint 2 Aug 15 – Aug 24 Goal: a user can find a match, book a session and leave feedback	Pathirana	- Implement the trust-score Cloud Function and surface trust badges in the UI - Implement report/block and emergency escalation flow - Support demo build and release tagging in Git
	Perera	- Deliver high-fidelity screens for matching, booking and feedback flows - Apply the design system consistently across all four components - Prepare the demo screen flow and presentation visuals
	Wijekoon	- Implement the matching Cloud Function (locality + skill + availability + trust score) - Implement ranked results with match-reason explanation - Run Sprint 2 review; re-prioritize the backlog for Sprints 3–4
	Ranketh	- Implement the full booking lifecycle with conflict handling and cancellation - Implement reminders (FCM + scheduled Cloud Function) and the post-session feedback loop - Run usability testing; compile Scrum event evidence for the report

6. Git Branch & Commit Responsibility

Each member commits under their own GitHub account on their own feature branch, merged via pull request reviewed by at least one teammate. Commit history from Sprint 0 onward evidences individual contribution as required by the assignment brief.

Member	Branch(es)	Expected Commits
Pathirana P P D S S	feature/auth, feature/trust-score	Flutter auth screens, Firestore security rules, verification workflow, trust-score Cloud Function
Perera K K L A	feature/profiles, feature/design-system	Profile models and screens, accessibility widget library, shared theme, /design assets
Wijekoon W M V M	feature/matching, docs/backlog	Matching Cloud Function, discovery/search UI, product backlog and charter documents
Ranketh K A D	feature/scheduling, docs/scrum-artifacts	Booking lifecycle, calendar UI, notifications, feedback loop, sprint board exports and burndown data

7. Report Writing Assignment

Target length: 10–20 pages. Each member drafts their sections; the full report is reviewed together before submission.

Report Section	Owner	Est. Pages
Cover page, contents, project selection & justification	Wijekoon	2
Project Charter (vision, scope, objectives, constraints, stakeholders)	Wijekoon	2
Initial Product Backlog (epics, stories, estimates)	Wijekoon + Ranketh	2.5
Sprint 1 & 2 backlogs, plans and burndown charts	Ranketh	3
Scrum roles and responsibilities	Ranketh	1
Evidence of Scrum events (planning, stand-up, review, retro logs)	Ranketh (all contribute notes)	2.5
Technology stack and architecture diagram	Pathirana	1.5
Working increment demo (screenshots, feature walkthrough)	Pathirana + Perera	2
Repository link and per-member commit summary	Pathirana	1
Team collaboration and communication (PM tool evidence)	Perera	1
Appendix: AI use declaration (CLEAR framework)	Perera	0.5

8. Scrum Ceremony Responsibility

Ceremony	Frequency	Lead	Team's Role
Sprint Planning	Start of sprint	Ranketh facilitates; Wijekoon presents backlog	All members estimate and commit to tasks
Daily Stand-up	Daily, 10 min	Ranketh (Scrum Master)	Each member: yesterday / today / blockers
Sprint Review	End of sprint	Pathirana & Perera demo; Wijekoon validates	Stakeholder feedback captured by Wijekoon
Retrospective	End of sprint	Ranketh (Scrum Master)	All members contribute improvements
Backlog Refinement	Mid-sprint	Wijekoon (Product Owner)	Team re-estimates upcoming stories

9. Notes

- Every member must be able to independently explain their component's design decisions, code and any AI-assisted content during the viva examination.
- Significant use of AI tools must be declared in the report appendix under the CLEAR framework (Collaborate, Learn, Evaluate, Acknowledge, Reflect).
- Sprints 3 and 4 fall outside the scope of Assignment 01 but the backlog is maintained for them.